Lesson 25: Convex Optimization Problems

AI/ML Mathematics

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1 Introduction

Optimization lies at the heart of nearly every algorithm in machine learning and reinforcement learning. Supervised learning reduces to empirical risk minimization; policy gradient methods maximize a surrogate objective at each iteration; and constrained Markov decision processes are formulated as saddle-point problems whose structure is governed by convexity and duality. A precise understanding of convex optimization problems—their standard form, the structure of their feasible sets, the classes of programs that arise in practice, and the fundamental theorem that local optimality implies global optimality—is therefore essential.

This lesson develops the theory of convex optimization problems from first principles. Section 2 introduces the standard form, the feasible set, and the optimal value. Section 3 proves that every local minimizer of a convex program is a global minimizer. Sections 4–7 present the principal classes of convex programs—linear programs, quadratic programs, second-order cone programs, and semidefinite programs—and Section 8 establishes the inclusion hierarchy among them. Section 9 works through detailed examples, and Section 11 provides exercises for further practice.

Throughout we assume familiarity with convex sets, convex functions, and their basic properties as developed in Lessons 23–24.

2 Standard Form of Convex Optimization

Definition 2.1 (Convex optimization problem)**.** A *convex optimization problem* in *standard form* is

$$\begin{aligned} \min_{x\in\mathbb{R}^n} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \le 0, \quad i = 1,\dots,m, \\ & a_j^\top x = b_j, \quad j = 1,\dots,p, \end{aligned} \tag{1}$$

where $f_0, f_1, \dots, f_m : \mathbb{R}^n \to \mathbb{R}$ are *convex* functions, $a_j \in \mathbb{R}^n$, and $b_j \in \mathbb{R}$. The function f_0 is called the *objective function*, the inequalities $f_i(x) \le 0$ are *inequality constraints*, and $a_j^\top x = b_j$ are *equality constraints*.

Remark 2.2. The equality constraints must be affine. If we were to allow a nonlinear equality constraint $h(x) = 0$ with h convex, the set $\{x : h(x) = 0\}$ would generally not be convex (it is convex only when h is affine). Requiring affine equality constraints ensures the feasible set is convex.

Definition 2.3 (Feasible set and optimal value)**.** The *feasible set* of (1) is

$$\mathcal{F} = \big\{x \in \mathbb{R}^n : f_i(x) \le 0 \text{ for } i = 1,\dots,m, \quad a_j^\top x = b_j \text{ for } j = 1,\dots,p\big\}. \tag{2}$$

The *optimal value* is

$$p^* = \inf\{f_0(x) : x \in \mathcal{F}\}. \tag{3}$$

We adopt the convention $p^* = +\infty$ if $\mathcal{F} = \emptyset$ (the problem is *infeasible*) and $p^* = -\infty$ if the objective is unbounded below on $\mathcal{F}$.

Definition 2.4 (Optimal point)**.** A point $x^* \in \mathcal{F}$ is an *optimal point* (or *global minimizer*) if $f_0(x^*) = p^*$. The set of all optimal points is denoted X_{opt}. A point x^* is a *local minimizer* if there exists $\delta > 0$ such that $f_0(x^*) \le f_0(x)$ for all $x \in \mathcal{F}$ with $\|x - x^*\| \le \delta$.

Proposition 2.5 (Convexity of the feasible set)**.** *The feasible set $\mathcal{F}$ of a convex optimization problem (1) is a convex set.*

Proof. Let $x, y \in \mathcal{F}$ and $t \in [0,1]$. Set $z = (1-t)x + ty$. For each inequality constraint, by convexity of f_i:

$$f_i(z) \leq (1-t)f_i(x) + tf_i(y) \leq (1-t) \cdot 0 + t \cdot 0 = 0.$$

For each equality constraint, by linearity:

$$a_j^\top z = (1-t)a_j^\top x + ta_j^\top y = (1-t)b_j + tb_j = b_j.$$

Hence $z \in \mathcal{F}$, so $\mathcal{F}$ is convex. □

Proposition 2.6 (Convexity of the optimal set)**.** *If $X_{opt} \neq \emptyset$, then X_{opt} is a convex set.*

Proof. Let $x^*, y^* \in X_{\text{opt}}$, so $f_0(x^*) = f_0(y^*) = p^*$. For $t \in [0,1]$, set $z = (1-t)x^* + ty^*$. By Proposition 2.5, $z \in \mathcal{F}$. By convexity of f_0:

$$f_0(z) \leq (1-t)f_0(x^*) + tf_0(y^*) = (1-t)p^* + tp^* = p^*.$$

Since z is feasible and p^* is the infimum, $f_0(z) \geq p^*$. Hence $f_0(z) = p^*$, so $z \in X_{\text{opt}}$. □

3 Local Optimality Implies Global Optimality

The following theorem is the most fundamental property of convex optimization. It asserts that every local minimum is automatically a global minimum, so there are no suboptimal local minima to get trapped in.

Theorem 3.1 (Local optimality implies global optimality)**.** *For the convex program (1), every local minimizer is a global minimizer. That is, if x^* is a local minimizer, then $f_0(x^*) = p^*$.*

Proof. Let x^* be a local minimizer: there exists $\delta > 0$ such that $f_0(x^*) \leq f_0(x)$ for all $x \in \mathcal{F}$ with $\|x - x^*\| \leq \delta$. Suppose, for the sake of contradiction, that there exists a feasible point $y \in \mathcal{F}$ with $f_0(y) < f_0(x^*)$.

Define the convex combination

$$z_t = (1-t)x^* + ty, \quad t \in (0,1). \tag{4}$$

Step 1: z_t is feasible. By Proposition 2.5, $\mathcal{F}$ is convex, so $z_t \in \mathcal{F}$ for all $t \in [0,1]$.

Step 2: $f_0(z_t) < f_0(x^*)$. By convexity of f_0:

$$f_0(z_t) \leq (1-t)f_0(x^*) + tf_0(y) < (1-t)f_0(x^*) + tf_0(x^*) = f_0(x^*), \tag{5}$$

where the strict inequality uses $f_0(y) < f_0(x^*)$ and $t > 0$.

Step 3: z_t is within δ of x^*. We have

$$\|z_t - x^*\| = t\|y - x^*\|.$$

Choose $t_0 \in (0,1)$ small enough that $t_0\|y - x^*\| < \delta$.

Contradiction. For $t = t_0$, the point z_{t_0} is feasible, satisfies $\|z_{t_0} - x^*\| < \delta$, and $f_0(z_{t_0}) < f_0(x^*)$. This contradicts the local optimality of x^*. □

Local Implies Global for Convex Problems

Every local minimizer of a convex optimization problem is a global minimizer. This is the principal reason convex problems are considered "tractable": any descent algorithm that converges to a local minimum has necessarily found the global minimum.

Corollary 3.2 (First-order optimality for unconstrained convex problems)**.** *If $f_0 : \mathbb{R}^n \to \mathbb{R}$ is convex and differentiable, then x^* is a global minimizer of f_0 if and only if $\nabla f_0(x^*) = 0$.*

Proof. ($\Leftarrow$) By convexity, for all $x \in \mathbb{R}^n$: $f_0(x) \geq f_0(x^*) + \nabla f_0(x^*)^\top (x - x^*) = f_0(x^*)$, so x^* is a global minimizer.

($\Rightarrow$) If $\nabla f_0(x^*) \neq 0$, then $d = -\nabla f_0(x^*)$ is a descent direction: $f_0(x^* + \epsilon d) < f_0(x^*)$ for small $\epsilon > 0$, contradicting global minimality. □

Warning

The theorem relies crucially on the convexity of both the objective f_0 and the feasible set $\mathcal{F}$. For non-convex problems, local minima may be strictly worse than the global minimum; this is the fundamental difficulty in training deep neural networks.

4 Linear Programs (LP)

Definition 4.1 (Linear program)**.** A *linear program* (LP) has the form

$$\min_{x \in \mathbb{R}^n} c^\top x \quad \text{s.t.} \quad Ax \leq b, \;\; Cx = d, \tag{6}$$

where $c \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{p \times n}$, $d \in \mathbb{R}^p$.

Remark 4.2. Both the objective $c^\top x$ and the constraints $A_i^\top x - b_i \leq 0$ are affine, hence convex. The feasible set is a *polyhedron*. By Theorem 3.1, every local optimum of an LP is a global optimum.

Example 4.3 (LP for MDP)**.** Given a discounted MDP with states $\mathcal{S}$, actions $\mathcal{A}$, transition probabilities $P(s'|s,a)$, rewards $r(s,a)$, and discount factor $\gamma \in (0,1)$, the value function satisfies the linear program:

$$\min_V \sum_{s \in \mathcal{S}} c(s) V(s) \quad \text{s.t.} \quad V(s) \geq r(s,a) + \gamma \sum_{s'} P(s'|s,a) V(s'), \quad \forall\, s \in \mathcal{S},\; a \in \mathcal{A},$$

where $c(s) > 0$ are state relevance weights. The constraints enforce the Bellman inequality, and the minimization pushes V down to the fixed point.

5 Quadratic Programs (QP)

Definition 5.1 (Quadratic program)**.** A *quadratic program* (QP) has the form

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^\top P x + q^\top x + r \quad \text{s.t.} \quad Ax \leq b, \;\; Cx = d, \tag{7}$$

where $P \in \mathbb{R}^{n \times n}$ is symmetric positive semidefinite ($P \succeq 0$), $q \in \mathbb{R}^n$, and $r \in \mathbb{R}$.

Remark 5.2. Setting $P = 0$ in (7) recovers a linear program. Hence LP $\subseteq$ QP.

Example 5.3 (Least-squares regression)**.** Given data matrix $A \in \mathbb{R}^{m \times n}$ and target $b \in \mathbb{R}^m$, the ordinary least-squares problem is

$$\min_x \frac{1}{2} \|Ax - b\|^2 = \frac{1}{2} x^\top (A^\top A) x - b^\top A x + \frac{1}{2} \|b\|^2,$$

which is a QP with $P = A^\top A \succeq 0$.

Example 5.4 (Quadratically constrained QP – QCQP)**.** A *QCQP* allows convex quadratic constraints:

$$\min_x \frac{1}{2} x^\top P_0 x + q_0^\top x \quad \text{s.t.} \quad \frac{1}{2} x^\top P_i x + q_i^\top x + r_i \leq 0, \quad i = 1, \ldots, m,$$

with each $P_i \succeq 0$. QCQPs arise in robust optimization and trust-region methods, where the trust region $\|x - x_k\|^2 \leq \Delta^2$ is a quadratic constraint.

6 Second-Order Cone Programs (SOCP)

Definition 6.1 (Second-order cone). The *second-order cone* (or *Lorentz cone*) in $\mathbb{R}^{n+1}$ is

$$\mathcal{K}_{n+1} = \{(x,t) \in \mathbb{R}^n \times \mathbb{R} : \|x\|_2 \le t\}.$$

Definition 6.2 (Second-order cone program). A *second-order cone program* (SOCP) has the form

$$\min_{x\in\mathbb{R}^n} f^\top x \quad \text{s.t.} \quad \|A_i x + b_i\|_2 \le c_i^\top x + d_i, \quad i = 1,\dots,m, \tag{8}$$

where $A_i \in \mathbb{R}^{n_i\times n}$, $b_i \in \mathbb{R}^{n_i}$, $c_i \in \mathbb{R}^n$, $d_i \in \mathbb{R}$.

Proposition 6.3 (SOCP constraints are convex). *Each constraint $\|A_i x+b_i\|_2 \le c_i^\top x+d_i$ defines a convex set.*

Proof. Define $g(x) = \|A_i x + b_i\|_2 - (c_i^\top x + d_i)$. The function $x \mapsto \|A_i x + b_i\|_2$ is convex as the composition of a norm (convex) with an affine map (which preserves convexity). Subtracting the affine function $c_i^\top x + d_i$ preserves convexity. Hence g is convex, and $\{x : g(x) \le 0\}$ is a convex set. □

Proposition 6.4 (QP embeds in SOCP). *Every quadratic program can be written as a second-order cone program. Hence QP $\subseteq$ SOCP.*

Proof. Consider the QP constraint $\frac{1}{2}x^\top Px + q^\top x + r \le 0$ with $P = R^\top R$ (Cholesky). Introduce t and rewrite as:

$$q^\top x + r \le t, \quad \|Rx\|_2 \le t,$$

using the fact that $\frac{1}{2}x^\top Px \le t$ for suitable t can be encoded via the rotated second-order cone. The linear objective is trivially an SOCP objective. □

Example 6.5 (Robust least-squares). When the data matrix A has uncertain rows, the robust formulation $\min_x \max_{\|u_i\|\le\epsilon} \|(A+U)x - b\|_2$ reduces to an SOCP.

7 Semidefinite Programs (SDP)

Definition 7.1 (Linear matrix inequality). A *linear matrix inequality* (LMI) is a constraint of the form

$$F(x) = F_0 + \sum_{i=1}^n x_i F_i \succeq 0, \tag{9}$$

where $F_0, F_1, \dots, F_n \in \mathbb{R}^{k\times k}$ are given symmetric matrices and $\succeq 0$ denotes positive semidefiniteness.

Definition 7.2 (Semidefinite program). A *semidefinite program* (SDP) has the form

$$\min_{x\in\mathbb{R}^n} c^\top x \quad \text{s.t.} \quad F_0 + \sum_{i=1}^n x_i F_i \succeq 0, \quad Ax = b, \tag{10}$$

where $c \in \mathbb{R}^n$, $F_i \in \mathbb{R}^{k\times k}$ symmetric, $A \in \mathbb{R}^{p\times n}$, $b \in \mathbb{R}^p$.

Proposition 7.3 (LMI defines a convex set). *The set $\{x \in \mathbb{R}^n : F_0 + \sum_i x_i F_i \succeq 0\}$ is convex.*

Proof. Let x and y satisfy the LMI, and let $t \in [0,1]$. Then

$$F_0 + \sum_i ((1-t)x_i + ty_i)F_i = (1-t)\Big(F_0 + \sum_i x_i F_i\Big) + t\Big(F_0 + \sum_i y_i F_i\Big).$$

A convex combination of positive semidefinite matrices is positive semidefinite, so the constraint is satisfied at $(1-t)x + ty$. □

Proposition 7.4 (SOCP embeds in SDP)**.** *Every second-order cone constraint $\|u\|_2 \le t$ can be written as an LMI:*

$$\begin{pmatrix} tI & u \\ u^\top & t \end{pmatrix} \succeq 0. \tag{11}$$

Hence SOCP $\subseteq$ SDP.

Proof. The matrix in (11) is positive semidefinite if and only if $t \ge 0$ and $t \cdot t - u^\top u \ge 0$ (by the Schur complement condition), i.e., $\|u\|_2^2 \le t^2$ with $t \ge 0$, which is equivalent to $\|u\|_2 \le t$. □

Example 7.5 (Maximum eigenvalue minimization)**.** Given symmetric matrices $A_0, A_1, \ldots, A_n$, the problem

$$\min_{x \in \mathbb{R}^n} \lambda_{\max}\Big(A_0 + \sum_{i=1}^n x_i A_i\Big)$$

can be written as an SDP: min t subject to $tI - A_0 - \sum_i x_i A_i \succeq 0$.

8 The Hierarchy of Convex Programs

Theorem 8.1 (Inclusion hierarchy)**.** *The following strict inclusions hold among classes of convex programs:*

$$LP \subsetneq QP \subsetneq QCQP \subsetneq SOCP \subsetneq SDP \subsetneq \text{Convex Programs}. \tag{12}$$

Proof. The inclusions $\subseteq$ were established above:

- LP $\subseteq$ QP: set $P = 0$ (Remark 5.2).
- QP $\subseteq$ QCQP: a QP is a QCQP with linear constraints (set $P_i = 0$ in the constraints).
- QCQP $\subseteq$ SOCP: each convex quadratic constraint can be rewritten as a second-order cone constraint (Proposition 6.4).
- SOCP $\subseteq$ SDP: by the Schur complement encoding (Proposition 7.4).
- SDP $\subseteq$ Convex Programs: an SDP is a convex program because the LMI constraint defines a convex set (Proposition 7.3) and the objective is linear.

For strictness, we exhibit a member of each larger class not in the smaller class:

- min $\frac{1}{2}x^2$ is a QP but not an LP (objective is quadratic).
- min x s.t. $x^2 \le 1$ is a QCQP but not a standard QP (quadratic constraint).
- min x s.t. $\|(y, z)\|_2 \le x$ involves a cone in 3 dimensions that is not equivalent to any single quadratic constraint.
- The constraint $\lambda_{\max}(X) \le 1$ on a matrix variable is an LMI but cannot be written as finitely many SOC constraints in general.
- Geometric programs (after log-transformation) are convex but involve exponential cones, which are not LMIs.

□

Remark 8.2 (Computational complexity). Problems higher in the hierarchy are generally harder to solve:

Class	Interior-point cost per iteration	Typical solver
LP	$O(n^2)$	Simplex, barrier
QP	$O(n^3)$	Active set, barrier
SOCP	$O(n^3)$	Barrier
SDP	$O(n^3k + n^2k^2)$	Barrier

Here n is the number of variables and k is the size of the LMI.

9 Detailed Examples

Example 9.1 (Portfolio optimization as QP)**.** An investor allocates wealth among n assets with expected returns $\mu \in \mathbb{R}^n$ and covariance matrix $\Sigma \succ 0$. The mean-variance problem is:

$$\min_{w\in\mathbb{R}^n} \frac{1}{2} w^\top \Sigma w \quad \text{s.t.} \quad \mu^\top w \geq r_{\min}, \quad \mathbf{1}^\top w = 1, \quad w \geq 0,$$

where $r_{\min}$ is a target return. This is a QP with $P = \Sigma \succ 0$, so f_0 is strictly convex. By Theorem 3.1, any local minimizer is the unique global minimizer.

Example 9.2 (Minimum-norm point in polyhedron)**.** Given a polyhedron $\mathcal{P} = \{x : Ax \leq b\}$, the problem $\min_{x\in\mathcal{P}} \|x\|_2^2$ is a QP with $P = 2I \succ 0$. The solution is the point in $\mathcal{P}$ closest to the origin.

Example 9.3 (SDP relaxation of combinatorial optimization)**.** The MAX-CUT problem for a graph with weight matrix W can be relaxed from $\max_{x_i\in\{-1,+1\}} \frac{1}{4}\sum_{ij} W_{ij}(1-x_ix_j)$ to the SDP:

$$\max_{X\succeq 0} \frac{1}{4}\operatorname{tr}(LX) \quad \text{s.t.} \quad X_{ii} = 1,$$

where $L = \operatorname{diag}(W\mathbf{1}) - W$ is the graph Laplacian. The Goemans–Williamson rounding algorithm guarantees at least 0.878 of the optimal value.

RL/ML Connection

LP formulation of MDPs. As shown in Example 4.3, the value function of a discounted MDP solves a linear program. This LP has $|\mathcal{S}|$ variables and $|\mathcal{S}||\mathcal{A}|$ constraints, and its dual yields the state-action occupancy measure, which characterizes the optimal policy.
SDP relaxations. In multi-agent RL, computing Nash equilibria in certain classes of games can be relaxed to SDPs. Lyapunov-based safety certificates also lead to SDP feasibility problems, where one searches for a matrix $P \succ 0$ satisfying linear matrix inequalities derived from the system dynamics.

10 Equivalent Reformulations

Proposition 10.1 (Epigraph reformulation)**.** *The problem* $\min_x f_0(x)$ *subject to* $f_i(x) \leq 0$, $Ax = b$ *is equivalent to*

$$\min_{x,t} t \quad \text{s.t.} \quad f_0(x) \leq t, \quad f_i(x) \leq 0, \quad Ax = b.$$

This transforms a possibly nonlinear objective into a linear one at the cost of one additional variable and one additional constraint.

Proof. If (x^*, t^*) solves the epigraph problem, then $t^* = f_0(x^*)$ at optimality (otherwise we could decrease t^*), and the constraints on x^* match the original. Conversely, if x^* solves the original, then $(x^*, f_0(x^*))$ solves the epigraph formulation. □

Example 10.2 (Chebyshev center)**.** The *Chebyshev center* of a polyhedron $\mathcal{P} = \{x : a_i^\top x \le b_i\}$ is the center of the largest ball inscribed in $\mathcal{P}$:

$$\max_{x_c, r} \; r \quad \text{s.t.} \quad a_i^\top x_c + r\|a_i\|_2 \le b_i, \quad r \ge 0.$$

This is an LP in the variables (x_c, r).

11 Exercises

Exercise 11.1 (Uniqueness under strict convexity)**.** Suppose f_0 is *strictly convex* (i.e., $f_0((1-t)x + ty) < (1-t)f_0(x) + tf_0(y)$ for $x \ne y$ and $t \in (0,1)$). Prove that if an optimal point x^* exists, it is unique. *Hint:* Assume two distinct optima and use the definition of strict convexity to reach a contradiction.

Exercise 11.2 (Convex feasibility as optimization)**.** Show that the *feasibility problem*—determining whether $\mathcal{F} \ne \emptyset$ for a convex program—can be reformulated as a convex optimization problem. *Hint:* Minimize $\max_i f_i(x)$ subject to $Ax = b$.

Exercise 11.3 (LP relaxation tightness)**.** Consider a finite MDP with $|\mathcal{S}| = 3$ states and $|\mathcal{A}| = 2$ actions. Write the LP for the value function explicitly. Show that the LP optimal solution V^* satisfies the Bellman equation $V^*(s) = \max_a [r(s,a) + \gamma \sum_{s'} P(s'|s,a)V^*(s')]$ for all s.

Exercise 11.4 (SOCP from norm constraints)**.** Show that the problem $\min c^\top x$ subject to $\|x\|_2 \le 1$ is an SOCP. Write it in the standard form (8).

Exercise 11.5 (SDP feasibility for Lyapunov stability)**.** Given a matrix $A \in \mathbb{R}^{n\times n}$, show that the system $\dot{x} = Ax$ is stable (all trajectories converge to the origin) if and only if there exists $P \succ 0$ such that $A^\top P + PA \prec 0$. Explain why searching for such P is an SDP feasibility problem.

Exercise 11.6 (Hierarchy verification)**.** Give an explicit example of a convex optimization problem that is an SOCP but not a QCQP. *Hint:* Consider a constraint involving the sum of norms $\|A_1x + b_1\| + \|A_2x + b_2\| \le 1$.

Exercise 11.7 (Optimality set structure)**.** Let $f_0(x_1, x_2) = |x_1| + |x_2|$ and $\mathcal{F} = \{(x_1, x_2) : x_1 + x_2 = 1,\ x_1 \ge 0,\ x_2 \ge 0\}$. Find the optimal value p^* and the optimal set X_{opt}. Verify that X_{opt} is convex.

References